

SAP Integration Software Scope

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1. Purpose & Background

Welspun's pipe-testing stations currently record their results into SAP by hand. At the ultrasonic-testing stations, an operator counts the indications shown by the testing machine and types the details against each pipe number in SAP. At the Hydrotester station, results are entered directly into SAP, and the pressure graph is produced on a manual printer. This manual process is slow and prone to mistakes.

Faclon will build a SAP data-entry application, hosted on Welspun's IT network, that automatically captures each test result, lets the operator review it, and pushes it to SAP with the correct rules applied. For the Hydrotester, a companion graph application will replace the manual printer by capturing the pressure-versus-time curve digitally.

The SAP data-entry application will be hosted on the central server as part of the losense platform. The graph application will be hosted on a local IPC/Panel PC installed (New) in the Hydrotester station.

2. Scope Overview

The software covers three testing stations and two applications. IUT and FUT are the two ultrasonic-testing (UT) stations; the Hydrotester is the hydrostatic pressure-testing station. The SAP

Entry application runs at all three stations, and an additional Graph application runs at the Hydrotester.

Station	What the tool does
IUT	SAP Entry application - captures indications from the test report and pushes the result to SAP.
FUT	SAP Entry application - same as IUT.
Hydrotester	Graph application captures the pressure curve; SAP Entry application records the result in SAP.

3. Functional Requirements

3.1 IUT & FUT Stations (SAP Entry application)

The IUT and FUT stations work the same way, so their requirements are described together. At both stations, the operator tests a pipe, the application reads the results from the machine's report files, the operator reviews and classifies the pipe, and the details are pushed to SAP.

Production Order (PO) configuration

- The application shows a global dropdown of Production Order (PO) numbers, filled automatically from SAP.
- For each PO number, the operator sets which prove-up rule applies: MUT Prove-Up (manual ultrasonic re-check) and/or X-Ray Prove-Up.
- These settings are saved against the PO number and reused for every pipe on that order, because the prove-up requirement is the same for all pipes on a given PO.

Automatic capture from test reports

When a pipe finishes testing, the machine produces a PDF test report and a JPEG image of the indications. The application saves both, and a parser reads them to capture:

- Pipe number - from the PDF report.
- Test date - from the PDF report.
- Indication name and count - from the report.
- General remarks from the previous station - read from SAP; the operator can add to them.
- Testing remarks for this station - typed by the operator.

- Result - worked out automatically (see below).

Automatic result

The result is decided from the indications found and the prove-up settings saved for the selected PO:

- MUT Prove-Up - if an indication requires MUT prove-up under the selected PO.
- X-Ray Prove-Up - if an indication requires X-Ray prove-up under the selected PO.
- SAT - if no indications are found (the pipe is accepted).

Pipe classification

Before the entry is sent to SAP, the operator classifies the pipe as one of:

- Calibration pipe - and, for a calibration pipe, marks whether it passed.
- Repeatability-test pipe.
- Production (regular) pipe.

Retest handling: if the pipe is a retest, its number often carries an appended suffix such as _1, _2 or _3. The application removes this suffix so the correct base pipe number is used.

Sending the entry to SAP

- The operator reviews all captured fields and files, enters the checksum value, and presses Upload to push the entry to SAP.
- Regular (production) pipes go to the normal SAP endpoint. Calibration and repeatability pipes go to a separate SAP endpoint, and their record appears in a separate row of the SAP report.
- Regular pipe timestamp: the time the upload succeeds. If the upload only succeeds in a later shift (for example, after a failure), the test time from the PDF report is used instead.
- Calibration/repeatability pipe timestamp: always the test time recorded in the PDF report (agreed rule).

3.2 Hydrotester Station

The Hydrotester station uses two applications that work together: a Graph application that captures the pressure curve, and a SAP Entry application that records the result in SAP. Together they replace the existing direct-to-SAP entry and the manual graph printer.

Graph application

- Reads live pressure data from the Hydrotester's SCADA system over an OPC UA connection.
- When a test starts, it increases the pressure polling frequency and begins plotting two graphs:

- Standard graph – lower polling frequency, drawn as a square wave; the emphasis is on confirming the hold time.
- Actual graph – higher polling frequency; captures the true rise and fall of pressure.
- When the test finishes, it sends the pressure, hold time and date/time to the central server, saves both graph data files, and pushes them to the central server for the SAP Entry application.

SAP Entry application

- Creates a new entry from the data received from the Graph application. The entry is not uploaded to SAP yet.
- The pipe number is obtained in one of three ways, in order of preference:
 - a. Automatically from the MES scanner (QR) data.
 - b. If no scanner data is available, from a dropdown of pipe numbers passed by the previous station (read from SAP).
 - c. If still not available, typed in manually.
- The operator chooses which text fields to print on the graph image (Pipe No, Date/Time, Pressure, Holding Time – one further field to be confirmed).
- The operator enters the checksum value and presses Upload to push the entry to SAP.
- Every field on an entry can be edited, subject to the common edit rule in Section 4.

4. Common Behaviour (All Stations)

- New entry per test: the application creates a new entry every time a pipe is tested, including retests.
- Preview: the operator can preview all fields and files before uploading.
- Retry on failure – network / SAP-down / connectivity: the application retries every 5 minutes for up to 3 attempts. If it still fails, the entry is marked failed, and the operator presses Uploads again later.
- Retry on failure – previous-station entry missing: the application keeps retrying every 5 minutes until the previous-station entry appears (exact cadence to be confirmed with the SAP team).
- Shift end: 15 minutes before the shift ends, the application highlights every entry still waiting to upload because the previous-station entry is missing.
- Shift change: if a successful upload happens after the shift has changed, the SAP time uses the time from the file instead of the upload time.
- 4-hour calibration alert: if 4 hours pass after a calibration pipe without a new calibration, the application shows an alert.

- Editing an uploaded entry: all fields can be edited except the machine-measured values - indication counts, pressure and hold time.
- Status display: each entry shows its current status (for example: preview, uploaded, retrying, waiting for previous station). The FTP upload status is shown separately for each entry.
- Previous-station data: a dedicated tab lists the pipe numbers recently passed by the previous station, along with the general remarks entered there, so the operator can see which pipes are ready and read the previous remarks before testing.

5. Application Task List

Task	IUT	FUT	Hydro SAP	Hydro Graph	Remarks
SAP Entry Application					
Dropdown to select PO No + result type for that PO	Yes	Yes	–	–	<i>PO numbers fetched from SAP</i>
Create a new entry each time a pipe is tested (including retests)	Yes	Yes	Yes	–	<i>PDF + JPG file at IUT/FUT; two graphs at Hydrotester</i>
Auto-select the result based on the selected PO No	Yes	Yes	–	–	
Let the user mark the entry as calibration, repeatability or regular pipe	Yes	Yes	–	–	
For a calibration pipe, also let the user mark whether it passed	Yes	Yes	–	–	
Let the user add a remark	Yes	Yes	–	–	
Show the general remark from previous stations and let the user append to it	Yes	Yes	–	–	
Correct the pipe number for a retest (remove appended suffixes like _1, _2, _3)	Yes	Yes	–	–	
Auto-fill the pipe number from scanner (MES) data	–	–	Yes	–	
If scanner data is missing, offer a dropdown of pipe numbers from the previous station	–	–	Yes	–	<i>Pipe numbers fetched from SAP</i>
Allow manual entry of the pipe number	–	–	Yes	–	<i>If the pipe was also not scanned at the previous station</i>

Let the user choose the text fields to show on the graph	–	–	Yes	–	<i>Pipe No, Date/Time, Pressure, Holding Time, Temperature</i>
Let the user preview all fields and files	Yes	Yes	Yes	–	<i>PDF + JPG at IUT/FUT; two graphs at Hydrotester</i>
Take the user input for the Checksum field	Yes	Yes	Yes	–	
Push data to SAP when Upload is pressed, using the current timestamp	Yes	Yes	Yes	–	<i>Regular pipes only at IUT/FUT</i>
For a non-regular pipe, push to a separate SAP endpoint when Upload is pressed with timestamp from file	Yes	Yes	–	–	
If the push fails due to network issues, retry 3 times at 5-min intervals, then fail	Yes	Yes	Yes	–	<i>Upload must be pressed again later</i>
If the push fails because the previous-station entry is missing, retry every 5 min	Yes	Yes	Yes	–	<i>To be confirmed with the SAP team</i>
If the successful push happens after the shift has changed, use the file time instead of the push time	Yes	Yes	Yes	–	
15 minutes before shift end, highlight all entries pending upload due to a missing previous-station entry	Yes	Yes	Yes	–	
4 hours after a calibration pipe, if no new calibration is done, show an alert	Yes	Yes	–	–	
For an uploaded entry, allow editing of all fields except indication counts, pressure and hold time	Yes	Yes	Yes	–	
Display the status of each entry (preview, uploaded, retrying, waiting for previous station, etc.)	Yes	Yes	Yes	–	
Display the FTP file-upload status for each entry	Yes	Yes	Yes	–	
Add a tab that shows pipe-number entries from the previous station	Yes	Yes	Yes	–	
Graph Application (Hydrotester)					
Read data from the Hydrotester SCADA OPC UA server	–	–	–	Yes	
On test start, increase the pressure polling frequency and start plotting the actual pressure graph	–	–	–	Yes	

On test start, start plotting the standard pressure graph	-	-	-	Yes	
On test finish, push the pressure, holding time and date/time to the central server	-	-	-	Yes	
On test finish, dump both graphs' data to file and push them to the central server	-	-	-	Yes	

6. SAP Data Points

6.1 IUT / FUT - Regular pipe

Field pushed to SAP	Source
Pipe No	Report file
Checksum	User input
Indication type with count	Report file
Result	User input, with auto-selection
Remarks	User input
General remarks	Existing data from SAP + new user input
Date/Time	Upload time if uploaded in the same shift; test time if uploaded in a different shift

Read from SAP: PO numbers; General comments; Pipe numbers from the previous station

Files: PDF report; JPEG indication image 6

6.2 IUT / FUT - Calibration / Repeatability pipe

Field pushed to SAP	Source
Pipe No	Report file
Date/Time	Always the test time from the PDF report (agreed rule)
OK / Not OK	User input

Read from SAP: None

Files: PDF report; JPEG indication image

6.3 Hydrotester pipe

Field pushed to SAP	Source
Pipe No	Auto-selection /dropdown/user input
Checksum	User input
Pressure	From the Graph application
Hold time	From the Graph application
Remarks	User input
Date/Time	Upload time if uploaded in the same shift; test time if uploaded in a different shift

Read from SAP: Pipe numbers from the previous station

Files: Standard graph image; Actual graph image